

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CSE/IT/ BS-M101A/ 2024-25
2024
MATHEMATICS-IA

Full Marks: 70

Times: 3 Hours

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer all questions

5 × 2 = 10

- Examine the convergence of the improper integral $\int_0^1 \frac{1}{x^2} dx$. 2
- Evaluate $\int_0^\infty \sqrt{x} e^{-x^3} dx$. 2
- Examine if the subset $S = \{(x, y, z) | x + y + z = 1\}$ of R^3 is a subspace of R^3 . 2
- Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 10$ on $[2, 3]$. 2
- Find the local extrema of the function $f(x) = |x|$, $x \in [-1, 1]$ with proper justification. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any five questions

5 × 12 = 60

- (a) Find the volume of the solid generated by revolving one arch of $y = \sin x$ about x-axis. 4
 - (b) Test for convergence : $\int_0^\infty \frac{x^2}{4x^4 + 25} dx$. 4
 - (c) Show that $B(x, y) = \int_0^\infty \frac{t^{x-1}}{(1+t)^{x+y}} dt$. 4
- (a) Find the area of the surface generated by revolving about the y-axis the part of the asteroid $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ that lies in the first quadrant. 4
 - (b) Using Mean Value theorem prove that $\frac{\pi}{6} + \frac{\sqrt{3}}{15} < \sin^{-1} \left(\frac{3}{5} \right) < \frac{\pi}{6} + \frac{1}{8}$. 4
 - (c) State and prove Lagrange's Mean Value theorem. 4
- (a) For the matrix $A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$, find a matrix P such that $P^{-1}AP$ is a diagonal matrix. 6
 - (b) If λ is an eigen value of a non-singular matrix A , then show that $\frac{1}{\lambda}$ is an eigen value of A^{-1} . 3
 - (c) If for two square matrices A and B of same order, $AB = B$ and $BA = A$, then show that A and B are both idempotent matrices. 3

$A \cdot A^{-1} = I$

 - (a) Show that the vectors $(2, 1, 1)$, $(1, 2, 2)$, $(1, 1, 1)$ of R^3 are linearly dependent. 4
 - (b) Let $S = \{\alpha, \beta, \gamma\}$, $T = \{\alpha, \beta, \alpha + \beta + \gamma\}$ be subsets of a real vector space V . Show that $L(S) = L(T)$. 4
 - (c) If $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{pmatrix}$, show that A is invertible. Also find A^{-1} using elementary row operations. 4
- (a) State Maclaurin's theorem. Use it to expand $\cos x$ with Lagrange's form of remainder after n terms. 2+2
 - (b) Investigate the maxima and minima of $1 + 2\sin x + 3\cos^2 x$, $0 \leq x \leq \frac{\pi}{2}$. 4
 - (c) Evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2}\right)^x$. 4

11. (a) Determine the condition for which the system of equations

$$x + y + z = 1$$

$$x + 2y - z = b$$

$$5x + 7y + az = b^2$$

admits of (i) only one solution, (ii) no solution, (iii) infinitely many solutions.

(b) Prove that
$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = (a^3 + b^3 + c^3 - 3abc)^2.$$

(c) Reduce the matrix $\begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \end{pmatrix}$ in a row reduced echelon form and hence find its rank.

12. (a) If A and P be both $n \times n$ matrices and P be non-singular, then prove that A and $P^{-1}AP$ have same eigen values.

(b) $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Use Cayley-Hamilton theorem to find A^{100} .

(c) Show that the set $\{(0,1,1), (1,0,1), (1,1,0)\}$ is a basis of R^3 .

13. (a) Examine whether $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (x + y, x - y)$, $(x, y) \in R^2$ is a linear mapping. If T is linear, then find $\text{Ker}T$.

(b) Let $T: R^3 \rightarrow R^3$ be a linear mapping such that the matrix representation of T is $\begin{pmatrix} 2 & 3 & 1 \\ 3 & 2 & 2 \\ 1 & 2 & 4 \end{pmatrix}$ relative to the ordered basis $\{(1,0,0), (1,1,0), (1,1,1)\}$ of R^3 (domain) and $\{(1,0,0), (0,1,0), (0,0,1)\}$ of R^3 (co-domain). Then find T .

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Paper name: **CHEMISTRY**
 Full Marks: 70

Paper code: **BS-CH101**
 Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

1. Define the term electron gain enthalpy.
2. What do you mean by the zero-point energy of a particle in a 1-D box?
3. Explain why H_2O is liquid but H_2S is gas
4. What is an ambident nucleophile?
5. What do you understand by racemic mixture?

5x2=10

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

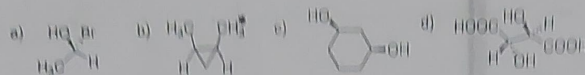
Answer any *four* questions

6. i) Prove that the energy associated with the motion of a particle in One dimensional box is quantized. 4x15 = 60
 Verify the uncertainty principle in one dimensional box. 3+2
 ii) Vitamin A has five conjugated double bonds and the first absorption peak at 332 nm. Find the length of the molecule. 3
 iii) Explain why the atoms or molecules show the continuous spectrum after ionization. 3
 iv) Consider a ball of mass 1gm moving with a speed of 1cm/s in one dimensional box of edge length equal to 10 cm. (a) Calculate the kinetic energy and number "n" corresponding to this kinetic energy. 4
 (b) If the ball is to be promoted to the next higher level, then how much of energy is required?
 7. i) What do you mean by effective nuclear charge? What would be the value of the screening constant according to Slater's rule for the valence electron of $_{33}As$ Calculate the electronegativity of $_{33}As$ (III). Given $r_{cov} = 120$ pm. 1+2+3
 ii) Define crystal field stabilization energy. Draw the splitting of d- orbitals under octahedral ligand field.
 iii) Explain why, complexes of Co(III) like $[Co(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$ and $[Co(NO_2)_6]^{3-}$ are orange-yellow in colour while $[CoF_6]^{3-}$, $[Co(H_2O)_6]^{3+}$ are blue in colour? 2+23
 iv) $[Fe(CN)_6]^{3-}$ is weakly paramagnetic whereas $[Fe(CN)_6]^{4-}$ is diamagnetic – explain. 2
 8. i) Derive the thermodynamic relation $\Delta S = C_v \ln T_2/T_1 + R \ln V_2/V_1$ for one mole of an ideal gas. (thermodynamic functions have their own significance). 3
 ii) Show that the decrease in the Gibbs free energy measures maximum available work under constant temperature and pressure. 2
 iii) Describe with proper example that the entropy of a system can be decreased during a spontaneous process and increased during a non – spontaneous process. 3
 iv) One litre of water at $25^\circ C$ and 1 atm pressure is subjected to a pressure of 100 atm, the constant temperature. Calculate the change in the Gibbs energy. 3
 v) Explain why C_p is always greater than C_v . Why is determining enthalpy change more useful than internal energy change in chemical reactions? 2+2
 9. i) Compare the ionization potential of nitrogen atom and nitrogen molecule based on the molecular orbital diagram. 4
 ii) Give the reason that $PbCl_2$ is white but PbI_2 is yellow in colour. 2
 iii) $AgCl$ is less soluble in water than $NaCl$ -why? 2

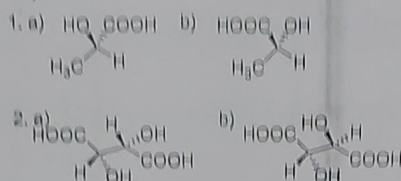
f f_{(x,y)} \cdot f_{(x,y)} = 0

- iv) Write two main differences between electron gain enthalpy and electronegativity. 2
 v) State the Lambert Beer Law? 1.25×10^{-4} M solution of a sugar solution in a 1 cm length cell at $\lambda_{max} = 197$ nm has absorbance 1.17. Calculate ϵ_{max} for this transition. 2+3

10. i) Identify the optically active molecule of the following! 4

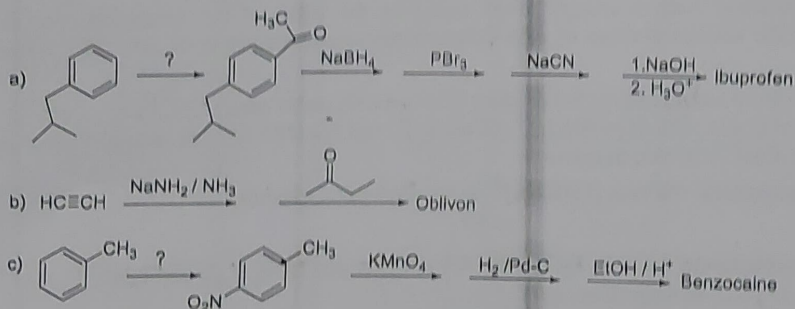


ii) Designate the chiral center as a (*R*) or (*S*) of the following and also identify the relationship (enantiomer, diastereomer or identical) 2+2



- iii) Calculate the specific rotation of a pure enantiomer of a given molecule, given that it has an optical rotation of $+0.52^\circ$ when measured in a 10 cm polarimeter tube at a concentration of 0.5 g per 10 ml. 4
 iv) Discuss the conformational isomers of n-butane. 3

11. i) Identify the product, when HBr was added to propene in presence of peroxide. 2
 ii) Explain the stability of the following carbocations: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2^+$, $(\text{CH}_3)_3\text{C}^+$, $\text{CH}_3\text{CH}_2\text{CH}^+\text{CH}_3$ 2
 iii) The reaction of alkyl chloride with aqueous KOH leads to the formation of alcohol; whereas alkenes are formed as major products in presence of alcoholic KOH. Explain. 3
 iii) Complete the following reactions and write the name of reagents where require:

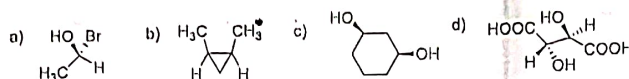


12. Write short notes on **any three** of the following: 3x5

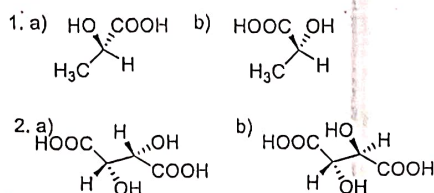
- (i) van der Waals forces of interaction (ii) n-type and p-type semiconductors (iii) Gibbs Helmholtz equation
 (iv) Mesomeric effect and hyperconjugation (v) Carbocation and carbanion (vi) Nucleophilic substitution

- iv) Write two main differences between electron gain enthalpy and electronegativity. 2
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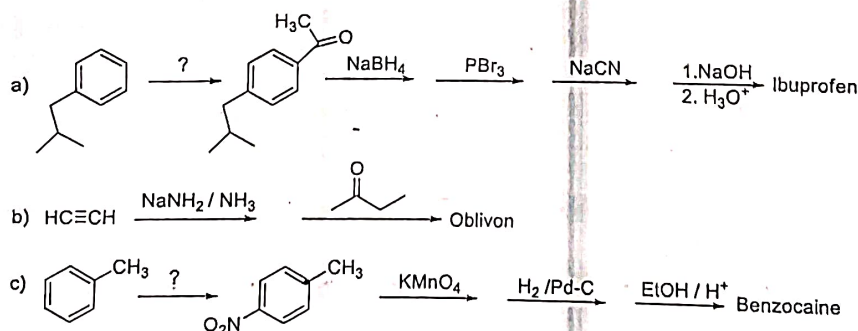
- iii) Calculate the specific rotation of a pure enantiomer of a given molecule, given that it has an optical rotation of $+0.52^\circ$ when measured in a 10 cm polarimeter tube at a concentration of 0.5 g per 10 ml. 4
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- (i) van der Waals forces of interaction (ii) n-type and p-type semiconductors (iii) Gibbs Helmholtz equation
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 2024
BASIC ELECTRICAL ENGINEERING

Times: 3 Hours

Marks: 70

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[OBJECTIVE TYPE QUESTIONS]

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5×2=10

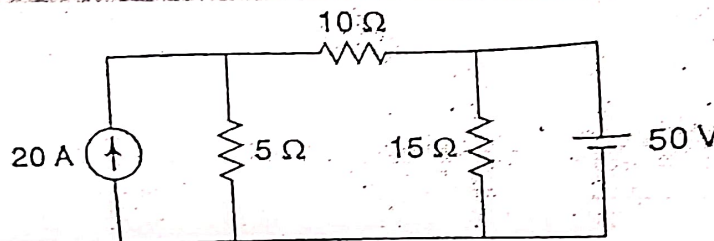
1. Differentiate between unilateral and bilateral circuit element. 2
2. Define form factor and peak factor of an alternating quantity. 2
3. Define Reluctance and permeance with their unit of a magnetic circuit. 2
4. Why the rating of transformer is given in KVA and not in KW? 2
5. Why single phase induction motor are not self-starting? 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

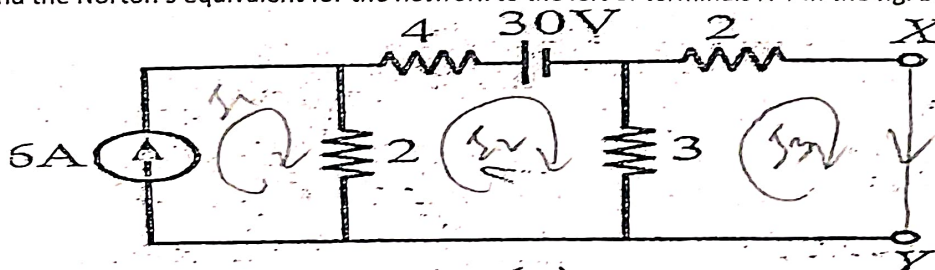
Answer any *four* questions

4×15 = 60

6. i) Define a linear circuit. Fully explain the superposition theorem for solving linear networks? 2+4
- ii) Find the current through 10 Ω resistor in fig. below using Thevenin's theorem. 5



ii) Find the Norton's equivalent for the network to the left of terminals X-Y in the fig. below- 4



7. i) What is the relationship between apparent power, true power and reactive power of an ac circuit? Draw the power triangle. 3
- ii) A resistance and inductance are connected in series across a voltage $v=283\sin 314t$. An expression of current is found to be $i=4\sin(314t-45^\circ)$. Find the values of resistance, inductance and power factor. 5
- iii) what is meant by resonance in series RLC Circuit? 2
- iv) A balanced star connected load of $(8+j6) \Omega$ per phase is connected to a balanced 3-phase, 400volt supply. Find the phase voltage, line current, power factor, power and total volt-ampere. 5

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8. i) What happens if dc supply is applied to the transformer? 2
 ii) In a 50 KVA, 1100/220 V transformer, the iron and copper losses at full load are 350 W and 425 W respectively. Calculate the efficiency at: 5
 a) full load with unity power factor. 4
 b) half load with unity power factor. 4
 iii) Derive the emf equation of an ideal single-phase transformer. 4
 iv) A 2000 KVA, 3300/240 volt, 50 Hz single-phase transformer has 80 turns on the secondary winding. Assuming an ideal transformer, calculate a) primary and secondary currents on full load b) the maximum value of flux and c) the number of primary turns. 4
9. i) Sketch a typical torque and slip curve of a 3-phase induction motor and explain. Does the Maximum torque depend on the rotor resistance per phase? 6
 ii) The power input to the rotor of a 400 Volt, 50 Hz, 6-pole, 3-phase induction motor is 80 kW. The rotor emf is observed to make 100 complete alternations per minute. Calculate slip, mechanical power developed and rotor copper loss. 4
 iii) A 3-phase induction motor operates from a supply whose frequency is 50 Hz and rotates at a speed of 1485 rpm at no load and 1350 at full load. Calculate: 5
 a) the speed at which the magnetic field of stator is rotating
 b) the slip at no load and at full load
 c) the frequency of the rotor current at no load and full load
 d) the frequency of rotor current at standstill.
10. i) Draw and explain the Speed-Torque characteristic of a Separately excited dc motor 4
 ii) A 4-pole, 220 volt dc separately excited motor has armature resistance of 0.2Ω . It takes 19 A at 220 V from the source while running at a speed of 1000 rpm. Find a) armature current 5
 b) back emf c) mechanical power developed. 2
 iii) Mention the purpose of commutators in DC motors? 4
 iv) Draw the connection diagram of a capacitor-start induction motor showing starting and main windings 5x3
11. Write short notes on any three of the followings:
- Miniature circuit breaker (MCB).
 - Lead-Acid Batteries.
 - Buck boost converter-working principle and application.
 - Comparison between cylindrical rotor and salient pole rotor.
 - Speed control of separately-excited Dc motor.
 - Single phase auto transformer.
 - fuse unit and its classifications.